Seminar 13 – CD 312 – 20.01.2022.

Ec. dif liniare de ordin n cu coef const.

$$a_0 \cdot y^{(n)} + a_1 \cdot y^{(n-1)} + \dots + a_{n-1} y' + a_n y = f(x).$$

? $y(x)$; $a_0, a_1, \dots, a_n \in \mathbb{R}$; $f(x)$ se cunosc.

– se caută o sol pt ec. omogenă $\Rightarrow y_0(x)$

– se caută o sol particulară pt ec neomog $\Rightarrow y_p(x)$.

$$\boxed{y(x) = y_0(x) + y_p(x)}$$

I) Ec. omogenă $a_0 \cdot y^{(n)} + a_1 y^{(n-1)} + \dots + a_{n-1} y' + a_n \cdot y = 0$. (*)

$n=1$: $a \cdot y' + b \cdot y = 0 \Rightarrow a \cdot y' = -by \Rightarrow \int \frac{y'}{y} = -\frac{b}{a} \Rightarrow \ln|y| = -\frac{b}{a}x + C$.

$y = k \cdot e^{\lambda x}$.

Euler: $n \in \mathbb{N}^*$ $\Rightarrow$ sol de forma $y = e^{\lambda x}$ pt EO (*)

$y'(x) = \lambda e^{\lambda x}$; $y'' = \lambda^2 e^{\lambda x}$, ..., $y^{(n)} = \lambda^n e^{\lambda x}$.

$a_0 \cdot \lambda^n e^{\lambda x} + a_1 \lambda^{n-1} e^{\lambda x} + \dots + a_{n-1} \lambda e^{\lambda x} + a_n e^{\lambda x} = 0$; $\forall x \in I$.

$$\boxed{a_0 \cdot \lambda^n + a_1 \cdot \lambda^{n-1} + \dots + a_{n-1} \cdot \lambda + a_n = 0.}$$

ec. caracteristică asoc ED.

(i) $\lambda_1, \lambda_2, \dots, \lambda_n$ rād reale distincte $\Rightarrow e^{\lambda_1 x}, e^{\lambda_2 x}, \dots, e^{\lambda_n x}$ sol lin ind. pt EDO.

$\Rightarrow$ sistem fundamental de soluții pt EO.

$$\boxed{y_{omog}(x) = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x} + \dots + C_n e^{\lambda_n x}}$$

(ii) $\lambda_1 = \lambda_2$; $\lambda_3, \dots, \lambda_n$ rād reale distincte $\Rightarrow e^{\lambda_3 x}, e^{\lambda_4 x}, \dots, e^{\lambda_n x}$ sol.

ER $\Rightarrow$ $e^{\lambda_1 x}$, $x \cdot e^{\lambda_1 x}$ sol $\Rightarrow$

$\Rightarrow y_{omg}(x) = C_1 e^{\lambda_1 x} + C_2 x e^{\lambda_1 x} + C_3 e^{\lambda_3 x} + \dots + C_n e^{\lambda_n x}$.

(iii) $\lambda_1 = a + ib$; $\lambda_3, \dots, \lambda_n \in \mathbb{R}$ distincte.

$\lambda_2 = a - ib$

$g(x) = e^{\lambda_1 x} = e^{(a+ib)x} = e^{ax} \cdot e^{ibx} = e^{ax}[\cos bx + i \sin bx]$.

$g(x) = e^{ax}\cos bx + i e^{ax} \sin bx \Rightarrow \mathrm{Re}\, g(x)$; $\mathrm{Im}\, g(x)$ sol reale pt ED.

$$\boxed{y_{omog}(x) = C_1 \cdot e^{ax}\cos bx + C_2 e^{ax} \sin bx + C_3 e^{\lambda_3 x} + \dots + C_n e^{\lambda_n x}.}$$

Ex 1 a) $y'' = 3y' - 2y$; b) $y'' + y = 0$; $y(0) = 2$; $y'(0) = 0$.
c) $y'' + 4y' + 4y = 0$; $y'(0) = 1$; $y(0) = 0$.

a) $y'' - 3y' + 2y = 0 \Rightarrow \lambda^2 - 3\lambda^1 + 2 = 0 \Rightarrow \lambda_{1,2} = <_2^1$.
e^x; e^{2x} sol pt ED $\Rightarrow$ $\boxed{y(x) = c_1 e^x + c_2 e^{2x}}$.

b) $y'' + y = 0 \Rightarrow \lambda^2 + 1 = 0 \Rightarrow \lambda_{1,2} = \pm i$
$e^{ix} = \cos x + i \sin x \Rightarrow y(x) = c_1 \cos x + c_2 \sin x$; $c_1, c_2 \in \mathbb{R}$
$y(0) = 2$; $y'(0) = 0$ cond inițiale CI (Cauchy).
$y'(x) = -c_1 \sin x + c_2 \cos x \Rightarrow y'(0) = c_2 = 0$.
$y(0) = c_1 = 2 \Rightarrow \boxed{y(x) = 2\cos x}$. sol unică.

c) $y'' + 4y' + 4y = 0$; $y(0) = 0$; $y'(0) = 1$.
$\lambda^2 + 4\lambda + 4 = 0 \Rightarrow (\lambda + 2)^2 = 0 \Rightarrow \lambda_1 = \lambda_2 = -2 \Rightarrow$
$\Rightarrow e^{-2x}, xe^{-2x}$ sol pt ED $\Rightarrow y(x) = c_1 e^{-2x} + c_2 x e^{-2x}$.
$y'(x) = -2c_1 e^{-2x} + c_2(e^{-2x} - 2xe^{-2x})$.
$y'(0) = -2c_1 + c_2$
$y(0) = c_1 + 0 = 0$ $\Rightarrow y'(0) = \boxed{1 = c_2} \Rightarrow \boxed{y(x) = xe^{-2x}}$.

Ec dif liniare neomogene cu coef const.

$a_0 \cdot y^{(n)} + a_1 \cdot y^{(n-1)} + \dots + a_{n-1} \cdot y' + a_n \cdot y = f(x)$.

2 metode. → metoda variației constantelor (Lagrange)
↳ ? sol particulară pt ec neomog de forma termen liber $f(x) \Rightarrow (y_p(x))$.
(dc f este o fct elem → exp; polinom; cos sau sin)

$\boxed{y(x) = y_0(x) + y_p(x)}$.

Cazuri particulare pt termu liber $f(x)$.

$y(x) = y_0(x) + y_p(x)$

1) $f(x) = Ae^{\alpha x} \Rightarrow$ (i) $y_p(x) = a \cdot e^{\alpha x}$ dc α nu e rād a ec caract.

⇓ (ii) $y_p(x) = a x^k e^{\alpha x}$ dc α e rād a ec caract cu mult k.

sol pt ec omog → $e^{\alpha x}; xe^{\alpha x}; \dots; x^{k-1}e^{\alpha x}$.

2) $f(x) =$ polinom de grad $m \Rightarrow y_p(x) =$ polinom de ac grad cu f.

3) $f(x) = a\cos\lambda x + b\sin\lambda x \Rightarrow y_p(x) = A\cos\lambda x + B\sin\lambda x$.

4) $f(x) = P_m(x)e^{\alpha x} \Rightarrow y_p(x) = x^k Q_m(x)e^{\alpha x}$ dc α e rād a ec. caract cu mult k.

Ex 2 a) $y'' - 3y' + 2y = e^{4x}$; b) $y'' - 3y' + 2y = e^{2x}$.

a) $y'' - 3y' + 2y = 0 \Rightarrow \lambda^2 - 3\lambda + 2 = 0 \Rightarrow \lambda_{1,2} = \begin{matrix}1\\2\end{matrix}$.

I) $\Rightarrow y_0(x) = c_1e^x + c_2e^{2x}$.

II) $f(x) = e^{4x} \Rightarrow y_p(x) = ae^{4x} \Rightarrow y_p' = 4ae^{4x}$, $y_p'' = 16ae^{4x}$.

$\Rightarrow 16ae^{4x} - 3\cdot 4ae^{4a} + 2\cdot ae^{4x} = e^{4x} \mid : e^{4x}$.

$16a - 12a + 2a = 1 \Rightarrow 6a = 1 \Rightarrow a = \frac{1}{6}$.

$$\boxed{y_p(x) = \frac{1}{6}e^{4x}}$$

III) $y(x) = y_0(x) + y_p(x) = c_1e^x + c_2e^{2x} + \frac{1}{6}e^{4x}$.

b) $y'' - 3y' + 2y = e^{2x}$. (*)

I) $y'' - 3y' + 2y = 0 \Rightarrow y_0(x) = c_1e^x + c_2e^{2x}$.

II) $f(x) = e^{2x} \Rightarrow y_p(x) = axe^{2x} \Rightarrow y_p' = ae^{2x}[1 + 2x]$.

$y_p'' = ae^{2x}[2 + 2(1+2x)] = ae^{2x}[4 + 4x]$. / $\Rightarrow$ (*)

$\Rightarrow ae^{2x}[4+4x] - 3ae^{2x}(1+2x) + 2axe^{2x} = e^{2x}$, $\forall x \Rightarrow (a = 1)$.

$y(x) = y_0(x) + y_p(x) = c_1 e^x + c_2 e^{2x} + xe^{2x}$.

Obs: $\begin{cases} y(0)=1. \\ y'(0)=2 \end{cases} \Rightarrow c_1, c_2$ se det in mod unic.

$\begin{cases} c_1 + c_2 = 1. \\ c_1 + 2c_2 + 1 = 2. \end{cases} \Rightarrow c_2 = 0. \quad c_1 = 1.$

Ex3 a) $y''' + y'' - y' - y = x^2 + 1$; b) $y''' + y'' = x + 1$.

c) $y'' - 5y' + 4y = \cos x$; d) $y'' - 3y' + 2y = xe^{3x}$.

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b)

$y_0(x) = c_1 + c_2 x + c_3 e^{-x}$.

$y_p(x) = x^2(ax+b) = ax^3 + bx^2$. $\Rightarrow a = \frac{1}{6}; b = 0$.

a) $y''' + y'' - y' - y = x^2 + 1$.

$\lambda^3 + \lambda^2 - \lambda - 1 = 0. \Rightarrow \lambda^2(\lambda+1) - (\lambda+1) = 0 \Rightarrow (\lambda^2-1)(\lambda+1) = 0$

$(\lambda-1)(\lambda+1)(\lambda+1) = 0.$

$\Rightarrow \begin{cases} \lambda_1 = \lambda_2 = -1 \\ \lambda_3 = 1. \end{cases}$

I) $y_0(x) = c_1 e^x + c_2 e^{-x} + c_3 xe^{-x}$.

II) $f(x) = x^2 + 1 \Rightarrow y_p(x) = ax^2 + bx + c$.

Metoda variației constantelor

1) $1 \cdot y^{(n)} + a_1(x) \cdot y^{(n-1)} + \dots + a_{n-1}(x) y' + a_n(x) y(x) = 0$.

$a_i(\cdot)$ cont.

Sol. gen $\Rightarrow$ $y_0(x) = C_1 \cdot y_1 + \dots + C_n y_n$

unde $y_1, \dots, y_n$ sol liniar ind.

2) $1 \cdot y^{(n)} + a_1(x) y^{(n-1)} + \dots + a_{n-1}(x) y' + a_n(x) y(x) = f(x)$

$\boxed{y(x) = c_1(x) \cdot y_1 + c_2(x) y_2 + \dots + c_n(x) \cdot y_n}$ unde

$c_1', \dots, c_n'$ verifică

$$\begin{cases} c_1' \cdot y_1 + \dots + c_n' y_n = 0 \\ c_1' \cdot y_1' + \dots + c_n' \cdot y_n' = 0 \\ \dots \\ c_1' \cdot y_1^{(n-1)} + \dots + c_n' \cdot y_n^{(n-1)} = f(x) \end{cases}$$

$\Downarrow$

$c_1(x) = \int c_1'(x)\,dx$

...

Ex 4 $\cdot y'' - 5y' + 4y = \cos x$.

I) $\lambda^2 - 5\lambda + 4 = 0 \Rightarrow \lambda_{1,2} = \begin{matrix} 1 \\ 4 \end{matrix} \Rightarrow y_0(x) = C_1 e^x + C_2 e^{4x}$.

II) $y(x) = c_1(x) e^x + c_2(x) e^{4x}$ unde $\left.\begin{matrix} y_1 = e^x \\ y_2 = e^{4x} \end{matrix}\right\}$.

c_1', c_2' verif $\begin{cases} c_1' \cdot y_1 + c_2' y_2 = 0 \\ c_1' \cdot y_1' + c_2' y_2' = f(x). \end{cases}$

$$\begin{cases} c_1' \cdot e^x + c_2' e^{4x} = 0. \quad (1) \\ c_1' \cdot e^x + c_2' \cdot 4e^{4x} = \cos x. \end{cases} \Big| \ominus \Rightarrow c_2' \cdot 3e^{4x} = \cos x.$$

$\Rightarrow c_2' = \frac{1}{3} e^{-4x} \cos x$. $\overset{(1)}{\Rightarrow} c_1' \cdot e^x = -c_2' \cdot e^{4x}$.

$c_1' = -e^{-x} \cdot \frac{1}{3} e^{-4x} \cos x \cdot e^{4x} = -e^{-x} \cos x$

$$\begin{cases} c_2(x) = \frac{1}{3} \int e^{-4x} \cos x\,dx = \dots \Rightarrow k_1 \\ c_1(x) = -\int e^{-x} \cos x\,dx = \dots \quad k_2. \end{cases}$$

$\boxed{y(x) = k_1 e^x + k_2 e^{4x} + \frac{3}{34} \cos x + \frac{5}{34} \sin x.}$